

Assignment - 5

Course: MST501-1: Python
Programming for Statistics



Done by:

Name: Ankita Das

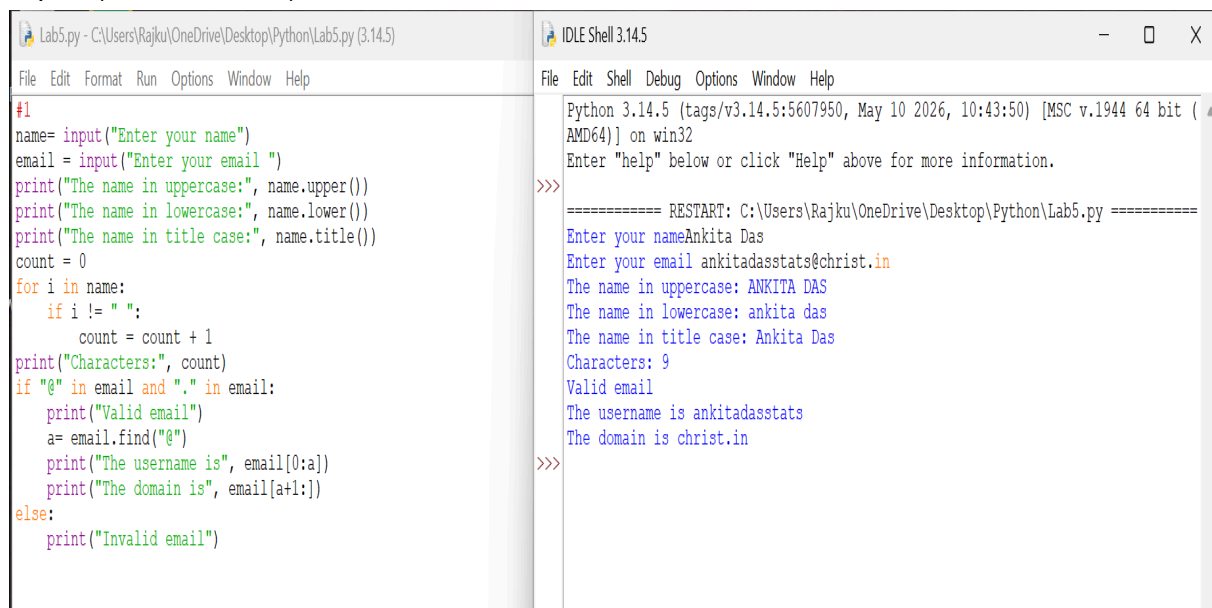
Register Number: 2648103

Date: 02/07/2026

1. Create a program that takes a student's full name and school email address as input. Display the name in uppercase, lowercase, and title case. Count the number of characters in the name (excluding spaces). Check whether the email contains both "@" and ".", and display the username and domain if the email is valid.

Ans-

```
name= input("Enter your name")
email = input("Enter your email ")
print("The name in uppercase:", name.upper())
print("The name in lowercase:", name.lower())
print("The name in title case:", name.title())
count = 0
for i in name:
    if i != " ":
        count = count + 1
print("Characters:", count)
if "@" in email and "." in email:
    print("Valid email")
    a= email.find("@")
    print("The username is", email[0:a])
    print("The domain is", email[a+1:])
else:
    print("Invalid email")
```



The screenshot shows a Python IDE with two panes. The left pane displays the source code of a program that takes a name and email as input, processes them, and prints the results. The right pane shows the output of the program, which includes the name in uppercase, lowercase, and title case, the character count, and the email validation results.

```
Lab5.py - C:\Users\Rajku\OneDrive\Desktop\Python\Lab5.py (3.14.5)
File Edit Format Run Options Window Help

#1
name= input("Enter your name")
email = input("Enter your email ")
print("The name in uppercase:", name.upper())
print("The name in lowercase:", name.lower())
print("The name in title case:", name.title())
count = 0
for i in name:
    if i != " ":
        count = count + 1
print("Characters:", count)
if "@" in email and "." in email:
    print("Valid email")
    a= email.find("@")
    print("The username is", email[0:a])
    print("The domain is", email[a+1:])
else:
    print("Invalid email")

IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab5.py =====
Enter your nameAnkita Das
Enter your email ankitadasstats@christ.in
The name in uppercase: ANKITA DAS
The name in lowercase: ankita das
The name in title case: Ankita Das
Characters: 9
Valid email
The username is ankitadasstats
The domain is christ.in
>>>
```

2. Create a program that takes a password and a sentence as input. Check whether the password is strong by verifying that it has at least 8 characters, one uppercase letter, one lowercase letter, one digit, and one special character. Then convert the sentence to lowercase, split it into words, and display how many times each word appears.

Ans-

```
password= input("Enter the password")
sentence= input("Enter the sentence")
```

```

u = 0
l = 0
d = 0
s = 0
for i in password:
    if i >= "A" and i <= "Z":
        u = 1
    elif i >= "a" and i <= "z":
        l = 1
    elif i >= "0" and i <= "9":
        d = 1
    else:
        s = 1
if len(password)>= 8 and u == 1 and l == 1 and d == 1 and s == 1:
    print("Strong Password")
else:
    print("Weak Password")
sentence = sentence.lower()
words = sentence.split()
for i in words:
    print(i, words.count(i))

```

```

#2
password= input("Enter the password")
sentence= input("Enter the sentence")
u = 0
l = 0
d = 0
s = 0
for i in password:
    if i >= "A" and i <= "Z":
        u = 1
    elif i >= "a" and i <= "z":
        l = 1
    elif i >= "0" and i <= "9":
        d = 1
    else:
        s = 1
if len(password)>= 8 and u == 1 and l == 1 and d == 1 and s == 1:
    print("Strong Password")
else:
    print("Weak Password")
sentence = sentence.lower()
words = sentence.split()
for i in words:
    print(i, words.count(i))

```

```

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab5.py =====
Enter the password123$Ablo
Enter the sentenceThis is my password and it must be a strong one
Strong Password
this 1
is 1
my 1
password 1
and 1
it 1
must 1
be 1
a 1
strong 1
one 1
>>>

```

3. Create a program that takes a chat message as input. First, remove extra spaces and replace common short forms such as "u" with "you" and "r" with "are". Then create a secret version of the cleaned message by shifting each alphabet letter forward by one (A→B, B→C, ..., Z→A). Finally, display both the cleaned message and the encoded secret message.

Ans-

```

message = input("Enter the message")
message =message.replace(" u ", " you ")

```

```

message=message.replace(" r ", " are ")
new=""
prev=""
for i in message:
    if not (i == " " and prev == " "):
        new= new+ i
    prev = i
print("The proper message", new)
sm=""
a="abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ"
b="bcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZA"
for i in new:
    found = 0
    for j in range(len(a)):
        if i== a[j]:
            sm = sm + b[j]
            found = 1
    if found== 0:
        sm =sm + i
print("The encoded secret message is", sm)

```

The screenshot shows a Python IDE with two windows. The left window, titled 'Lab5.py - C:\Users\Rajku\OneDrive\Desktop\Python\Lab5.py (3.14.5)', contains the following code:

```

"""
#3
message = input("Enter the message")
message=message.replace(" u ", " you ")
message=message.replace(" r ", " are ")
new=""
prev=""
for i in message:
    if not (i== " " and prev == " "):
        new= new+ i
    prev = i
print("The proper message", new)
sm=""
a="abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ"
b="bcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZA"
for i in new:
    found =0
    for j in range(len(a)):
        if i== a[j]:
            sm = sm + b[j]
            found = 1
    if found== 0:
        sm =sm + i
print("The encoded secret message is", sm)

```

The right window, titled 'IDLE Shell 3.14.5', shows the execution output:

```

Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab5.py =====
Enter the messageHow r u ?
The proper message How are you ?
The encoded secret message is Ipx bsf zpv ?
>>>

```